

Data Preprocessing & EDA

Week 3 - Day 2

The Art of Feature Engineering and Data Augmentation

Today, we journey beyond just cleaning datasets—we'll **engineer features** that supercharge our models and **augment data** where quantity is scarce. Imagine transforming plain spreadsheets into powerful prediction tools—**that's the magic of today's session.**



What You'll Learn



Data Augmentation



Why preprocessing is critical for ML success?



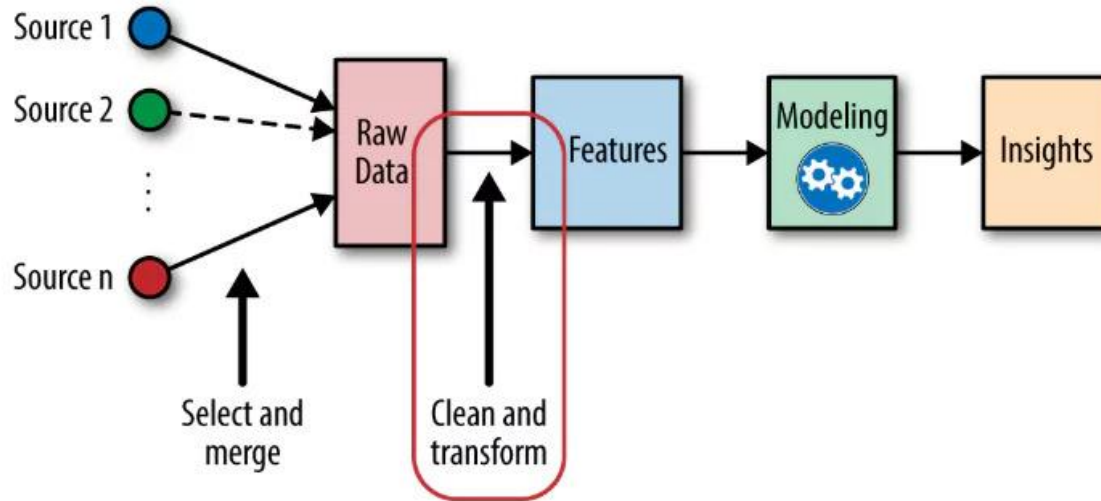
When Data Isn't Enough – Why Augment?

By the end of this section, you'll see preprocessing not as a chore, but as a _strategic step_ to unlock your model's potential."



What is Data Augmentation?

Data augmentation is a technique of artificially increasing the training set by creating modified copies of a dataset using existing data. It includes making minor changes to the dataset or using deep learning to generate new data points.



Augmented vs. synthetic data

1

Augmented data:

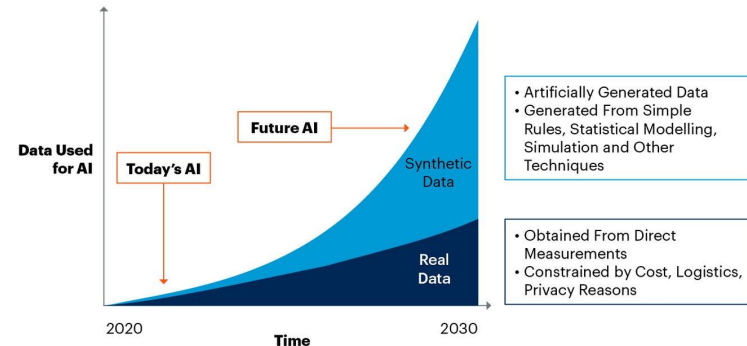
This involves creating modified versions of existing data to increase dataset diversity. For example, in image processing, applying transformations like rotations, flips, or color adjustments to existing images can help models generalize better.

2

Synthetic data:

This refers to artificially generated data, which allows researchers and developers to test and improve algorithms without risking the privacy or security of real-world data.

By 2030, Synthetic Data Will Completely Overshadow Real Data in AI Models



Source: Gartner
750175_C

Why is data augmentation important?

Data augmentation helps machine learning models perform better by making the most of existing data. It prevents overfitting, improves accuracy, and creates diversity in training data, which is crucial when datasets are small or imbalanced. By simulating real-world variations, it makes models more robust and reliable—without the need for expensive data collection. In short, it's a simple, powerful way to help models learn and generalize effectively.

DATA AUGMENTATION



When should you use data augmentation?

1

To prevent models from overfitting.

2

The initial training set is too small.

3

To improve the model accuracy & To Reduce the operational cost of labeling and cleaning the raw dataset



Data Augmentation Techniques

Audio Data Augmentation

Text Data Augmentation

Image Augmentation

Advanced Techniques

Audio Data Augmentation

1. **Noise injection:** add gaussian or random noise to the audio dataset to improve the model performance.
2. **Shifting:** shift audio left (fast forward) or right with random seconds.
3. **Changing the speed:** stretches times series by a fixed rate.
4. **Changing the pitch:** randomly change the pitch of the audio.

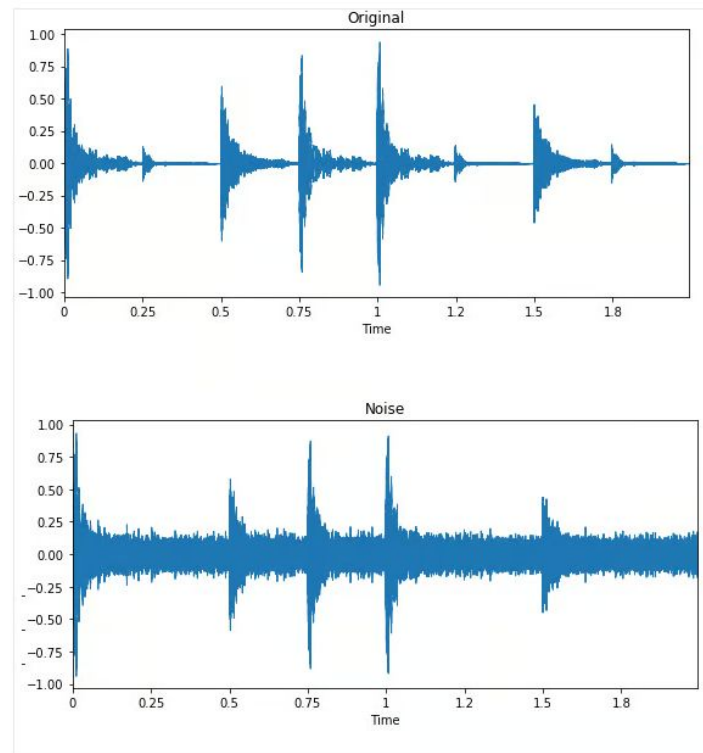


Image by [Edward Ma](#) | Noise Injection

Text Data Augmentation

1. **Word or sentence shuffling:** randomly changing the position of a word or sentence.
2. **Word replacement:** replace words with synonyms.
3. **Syntax-tree manipulation:** paraphrase the sentence using the same word.
4. **Random word insertion:** inserts words at random.
5. **Random word deletion:** deletes words at random.

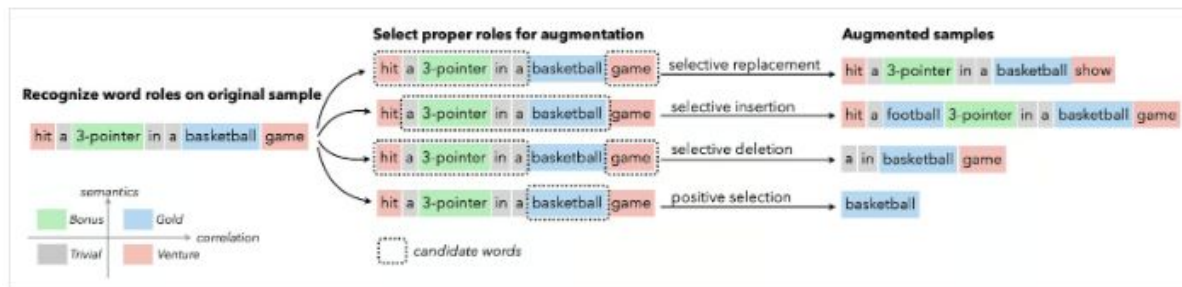
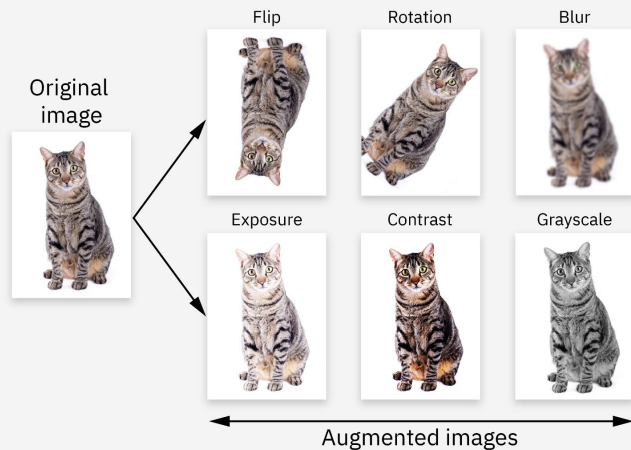


Image from [Papers With Code](#) | Selective Text Augmentation with Word Roles for Low-Resource Text Classification.

Image Augmentation

1. **Geometric transformations:** randomly flip, crop, rotate, stretch, and zoom images. You need to be careful about applying multiple transformations on the same images, as this can reduce model performance.
2. **Color space transformations:** randomly change RGB color channels, contrast, and brightness.
3. **Kernel filters:** randomly change the sharpness or blurring of the image.
4. **Random erasing:** delete some part of the initial image.
5. **Mixing images:** blending and mixing multiple images.

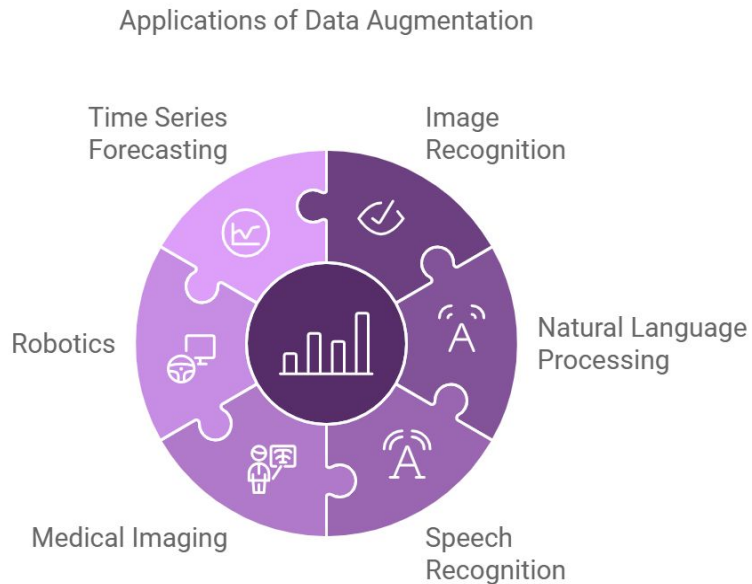


Advanced Techniques

1. **Generative adversarial networks** (GANs): used to generate new data points or images. It does not require existing data to generate synthetic data.
2. **Neural Style Transfer**: a series of convolutional layers trained to deconstruct images and separate context and style.

Data Augmentation Applications

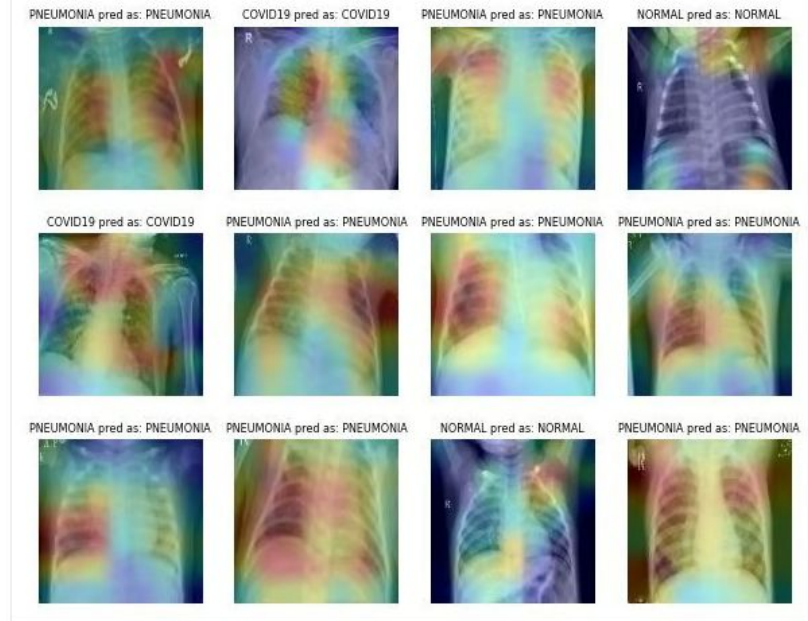
Data augmentation can apply to all machine learning applications where acquiring quality data is challenging. Furthermore, it can help improve model robustness and performance across all fields of study.



Healthcare

Acquiring and labeling medical imaging datasets is time-consuming and expensive. You also need a subject matter expert to validate the dataset before performing data analysis. Using geometric and other transformations can help you train robust and accurate machine-learning models.

For example, in the case of Pneumonia Classification, you can use random cropping, zooming, stretching, and color space transformation to improve the model performance. However, you need to be careful about certain augmentations as they can result in opposite results. For example, random rotation and reflection along the x-axis are not recommended for the X-ray imaging dataset.



Ethical Implications of Data Augmentation

While data augmentation is a powerful tool for enhancing machine learning models, it raises several ethical concerns that require careful consideration:

1. **Bias amplification:** Augmenting biased datasets can replicate and exacerbate existing inequities, leading to models that perform poorly for underrepresented groups.
2. **Privacy risks:** Synthetic data generation can unintentionally retain sensitive details from the original dataset, risking breaches of privacy regulations.
3. **Data authenticity:** Misapplied augmentations can produce unrealistic or misleading data points, potentially compromising model performance and trust.
4. **Transparency:** It is crucial to clearly document augmentation techniques and ensure stakeholders understand how data is being transformed or generated.
5. **Fairness and equity:** Ensuring augmented data reflects diverse populations and scenarios is essential to avoid unintended biases and ensure robust model performance.
6. **Regulatory compliance:** Augmented datasets must comply with data protection laws and ethical guidelines to avoid legal and reputational risks.

To responsibly leverage data augmentation, practitioners should validate augmented data, address biases, and ensure compliance with relevant ethical and legal standards.

Data Augmentation with Keras

Sequential

Resize and rescale

```
IMG_SIZE = 180

resize_and_rescale = keras.Sequential([
    layers.Resizing(IMG_SIZE, IMG_SIZE),
    layers.Rescaling(1./255)
])

result = resize_and_rescale(image)
plt.axis('off')
plt.imshow(result);
```

Random rotate and flip

```
data_augmentation = keras.Sequential([
    layers.RandomFlip("horizontal_and_vertical"),
    layers.RandomRotation(0.4),
])

plt.figure(figsize=(8, 7))
for i in range(6):
    augmented_image = data_augmentation(image)
    ax = plt.subplot(2, 3, i + 1)
    plt.imshow(augmented_image.numpy()/255)
    plt.axis("off")
```

Data Augmentation using tf.image

Flip left to right

```
flipped = tf.image.flip_left_right(image)  
visualize(image, flipped)
```

Original image



Augmented image



Grayscale

```
grayscaled = tf.image.rgb_to_grayscale(image)  
visualize(image, tf.squeeze(grayscaled))
```

Original image



Augmented image



Adjusting the saturation

```
saturated = tf.image.adjust_saturation(image, 3)  
visualize(image, saturated)
```

Original image



Augmented image



Adjusting the brightness

```
bright = tf.image.adjust_brightness(image, 0.4)  
visualize(image, bright)
```

Original image



Augmented image



Data Augmentation with Image DataGenerator

The Keras ImageDataGenerator is even simpler. It works best when you are loading data from a local directory or CSV.

```
datagen = keras.preprocessing.image.ImageDataGenerator(rotation_range=20,  
    width_shift_range=0.2,  
    height_shift_range=0.2,  
    horizontal_flip=True,  
    validation_split=0.2)  
  
datagen.fit(x_train)
```

Data Augmentation Tools

In this section, we will learn about other open-source tools that you can use to perform various data augmentation techniques and improve the model performance.

Pytorch

Image transformation is available in the [torchvision.transforms](#) module. Similar to Keras, you can add transform layers within `torch.nn.Sequential` or apply an augmentation function separately on the dataset.

Augmentor

[Augmentor](#) is a Python package for image augmentation and artificial image generation. You can perform Perspective Skewing, Elastic Distortions, Rotating, Shearing, Cropping, and Mirroring. Augmentor also comes with basic image pre-processing functionality.

Albumentations

[Albumentations](#) is a fast and flexible Python tool for image augmentation. It is widely used in machine learning competitions, industry, and research to improve the performance of deep convolutional neural networks.

Imgaug

Imgaug is an open-source tool for image augmentation. It supports a wide variety of augmentation techniques, such as Gaussian noise, contrast, sharpness, crop, affine, and flip. It has a simple yet powerful stochastic interface, and it comes with keypoints, bounding boxes, heatmaps, and segmentation maps.

OpenCV

OpenCV is a massive open-source library for computer vision, machine learning, and image processing. It is generally used in building real-time applications. You can use OpenCV to augment images and videos hassle-free.